

Total No. of Questions : 8]

SEAT No. :

P6756

[6181] - 344

[Total No. of Pages : 2

B.E. (Computer Engineering) (Honors)
VIRTUAL REALITY FOR GAME DEVELOPMENT
(2019 Pattern) (Semester - VII) (410701)

Time : 2 ½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.No.1 or Q.No.2, Q.No.3 or Q.No.4, Q.No.5 or Q.No.6, Q.No.7 or Q.No.8.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.

Q1) a) Draw and explain block diagram for a multimodal VR system with various software and hardware components. **[9]**

b) What is SMART stand for in define stage of iterative design. **[5]**

c) What different types of project constraints are in define stage? **[4]**

OR

Q2) a) Write short note on : **[9]**

i) Define Stage

ii) Make Stage

iii) Learn Stage

b) What different types of Project Constraints are in define stage of iterative design. **[5]**

c) Which are important factor during communication in learn stage. **[4]**

Q3) a) Explain working with Unity's UI System **[6]**

b) Write short note on Fundamentals of sprite animation **[6]**

c) What are different unity 2D feature? **[5]**

OR

P.T.O.

- Q4)** a) State comparison between Unity and Unreal Engine. [6]
b) Describe different steps to develop town view using unity. [6]
c) Write note on: NPCs and Interactions. [5]

- Q5)** a) How to create a battle system in VR environment. [9]
b) Explain workflow of Mecanim Animation System. [8]

OR

- Q6)** a) State different pre-processor directives that Unity recognizes. [9]
b) How to building in-game menu structures. Explain. [8]

- Q7)** a) Explain system delays in VR systems with diagram. [9]
b) Explain the terms w.r.t. physical issues involved with the use of VR equipment. [9]
i) Physical fatigue
ii) Headset fit
iii) Injury, and
iv) Hygiene

OR

- Q8)** a) Explain in details different theories of motion sickness. [9]
b) State system factor that contributes to adverse effects. [9]

